UNIGB TRAINING COURSE

Orchestrate Report Generation with AI

Lesson 1

n8n | Python | pptxgenjs | LLM AI

Automate, replicate, and orchestrate

by Nicola Bisinella - CTO & Innovation Manager - AI Enthusiastic - *©2026 - CC BY*

Lessons 1
Logistics

Total Duration: 12 hours across three workshop sessions.

Structure: 3 workshops (4 hours each).

Breaks: A 10-minute break will be taken every hour.

Q&A: Please feel free to interrupt and ask questions at any time.

Materials: Links to resources will be provided at the end of each session. Alternatively, you will receive an email with the link as soon as possible.

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- Digital Marketing Consultant

Keep in touch

Question 1

How many of you have already used ChatGPT or another generative AI tool for work, study, or in your personal life?

Question 2

Who has ever created or administered a survey (Google Forms, Typeform, paper-based...) and then spent hours manually organizing the data, maybe in Excel?

Question 3

Who thinks that AI and automation are stuff for IT people and don't really have much to do with tourism or food & wine management?

Course Agenda

Course Structure

01

What is n8n

Architecture, base components, fundamental nodes

02

Orchestrate informations

Raw Data → Processing → Agentic

03

RAW Data Processing

File structure, cleaning, key variables

04

The 3 Stages

Parse, Statistical Processing, PPTX Generation

05

Building the Workflow

n8n node architecture and design decisions

06

AI & Deploy

LLM API for commentary, QA, deployment

Lesson 1 Objectives

- Understand n8n Architecture (Canvas, Nodes, JSON)
- Map Prof. Pezzano’s Analysis Steps to Automated Nodes
- Build your first AI-Powered "Prompt Orchestrator"
- Learn to extract insights from Survey.xlsx files

What is n8n?

n8n is an **open-source workflow automation** platform that allows connecting apps, APIs, and services through a visual drag-and-drop interface with a **specific and reiterable process**.

Designed for technical & smart teams, it allows executing complex logic through **configurable nodes** — from HTTP Request to Code (Python/JS) all the way to AI Agent with LLM.

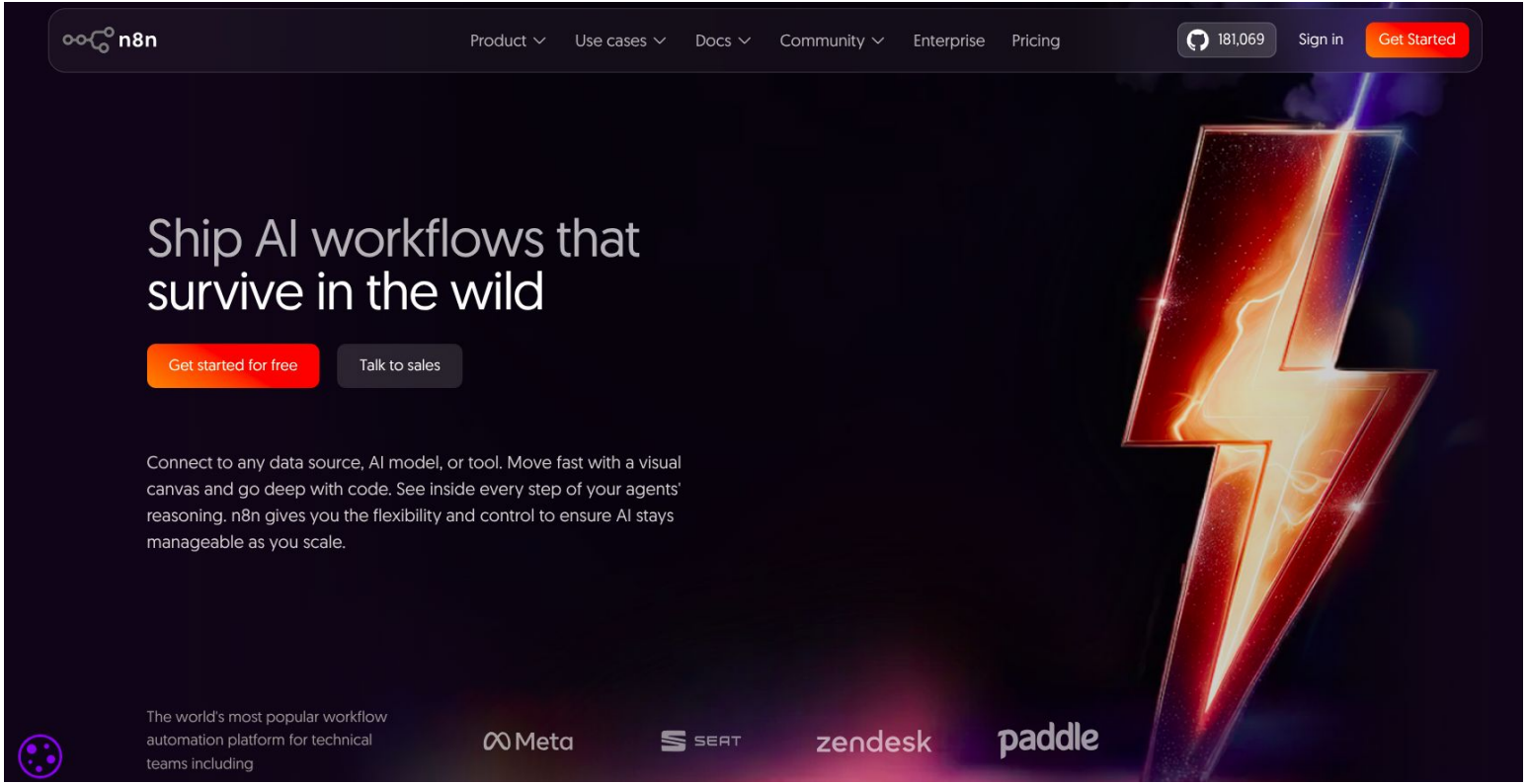
- Self-hosted: data always under your control
- 400+ native integrations ready to use
- Code nodes in JavaScript and Python
- Local or cloud execution (n8n.cloud)

400+
Integrations

OSS
Open Source

JS/Py
Code Nodes

AI
LLM Integration


n8n
Product
Use cases
Docs
Community
Enterprise
Pricing
181,069
Sign in
Get Started
Ship AI workflows that survive in the wild
Get started for free
Talk to sales
Connect to any data source, AI model, or tool. Move fast with a visual canvas and go deep with code. See inside every step of your agents' reasoning. n8n gives you the flexibility and control to ensure AI stays manageable as you scale.
The world's most popular workflow automation platform for technical teams including
Meta
SEAT
zendesk
paddle

https://n8n.io

Take a look into:

Quick demo

Workflow Components

TRIGGER

Starts the workflow (Manual, Webhook, Schedule, Form)

→

NODE

Processes data (Code, Set, HTTP Request)

→

NODE

Conditional logic (IF, Switch, Merge)

→

NODE

AI Action (AI Agent, HTTP Claude API)

→

OUTPUT

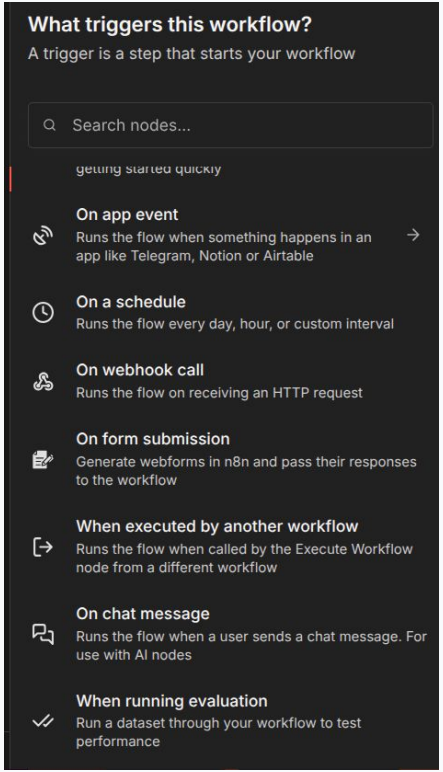
Final result (File, Email, Webhook, Slack)

Every workflow is composed of nodes connected in sequence. Data flows as JSON objects from one node to the next.

Workflow Components

TRIGGER

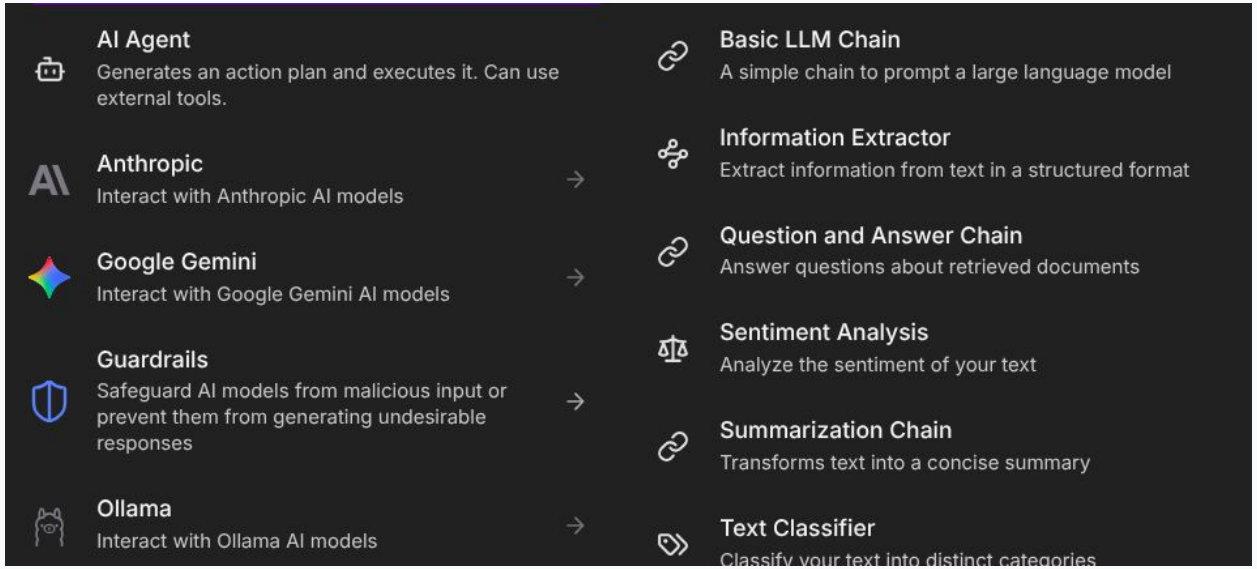
Starts the workflow (Manual, Webhook, Schedule, Form)



Workflow Components

NODE

AI Action
(AI Agent,
HTTP Claude API)



Exercise 1
“The AI Sentiment Butler”

30-45 min

- Register to free trial cloud account at https://n8n.io
- Create your first workflow
- **Objective**: Build a simple automation that takes a messy customer comment (e.g., from a survey) and automatically tells the manager if the customer is Happy or Angry, and what the next Management Action should be.
- **The Scenario**: You are a manager receiving hundreds of survey comments like: "The wine was great, but the waiter was slow and I hated the price."
  Reading them all takes hours. You want n8n to be your "Butler"—reading the comments and giving you a 2-second summary.

Exercise 1

“The AI Sentiment Butler”

Necessary data

OpenRouter Test API:
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

Use this QR code with your phone if you want copy the code quickly

Exercise 1

“The AI Sentiment Butler”

The 3-Step Build

Step 1: The Input (The "Letter")

1. Open n8n and click "Add Node".
2. Search for "When Chat Message Received" (Chat Trigger).
3. Concept: This node is the "Door." When someone (a customer or a manager) types something, the "Door" opens and the data enters n8n.

Step 2: The Brain (The "Analyst")

1. Connect the first node to a new node: "AI Agent".
2. In the System Message, tell the AI who it is:
   > "You are a Senior Manager. Analyze the customer comment. Identify if the sentiment is Positive, Neutral, or Negative. Then, provide ONE 'Action Item' for the staff (e.g., 'Give a discount' or 'Send a thank you note')."
3. In the Prompt field, use the data from the previous node:
   > "Analyze this: {{ $json.chatInput }}"
4. Concept: This node is the "Brain." It doesn't just copy text; it understands it and makes a decision based on your instructions.

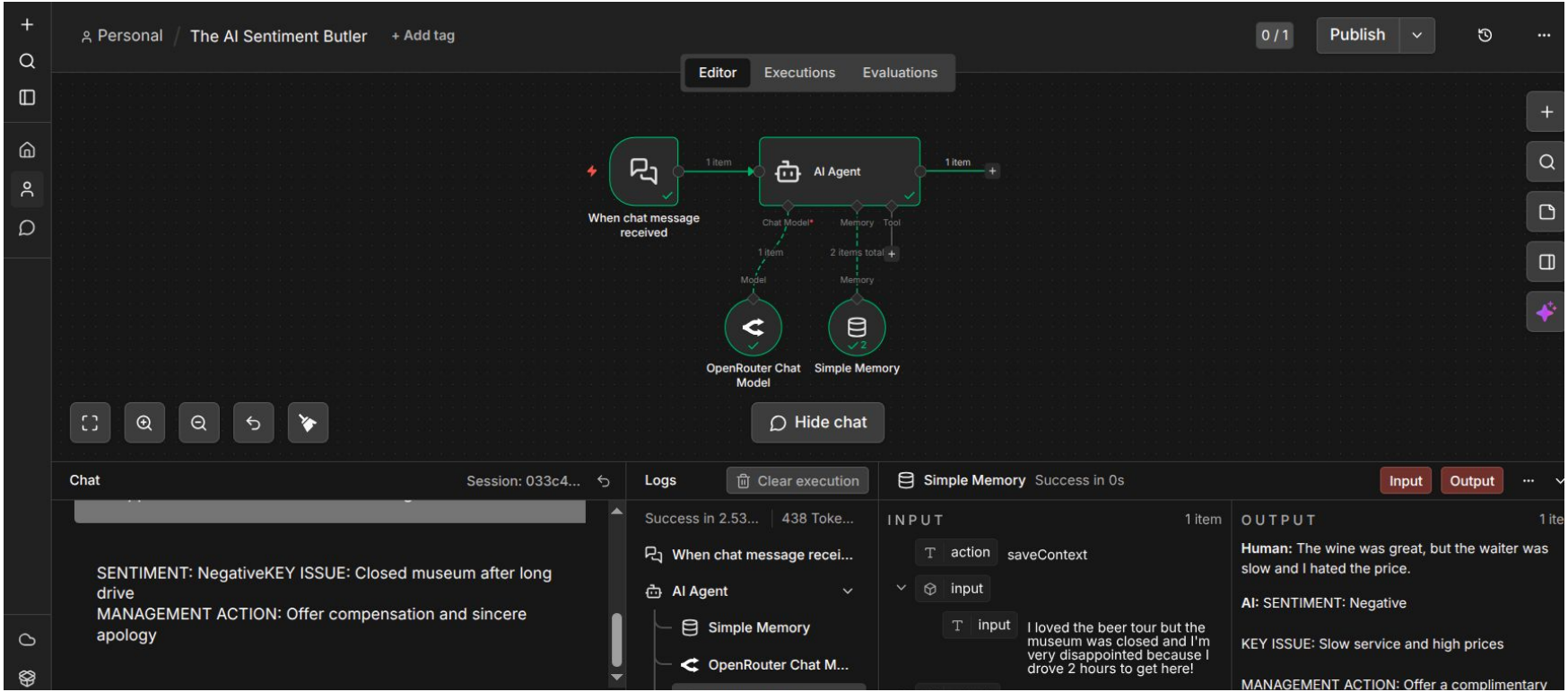
Step 3: The Output (The "Report")

1. Add the final node: "Respond to Webhook" (Chat Response).
2. In the Response field, type:
   > "Butler Report: {{ $json.output }}"
3. Concept: This is the "Result." It sends the AI's professional analysis back to you.

Exercise 1

“The AI Sentiment Butler”

It’s look like this?



Exercise 1

“The AI Sentiment Butler”

Testing

The "A-HA!" Moment

1. Click "Open Chat/Execute Workflow".
2. Open the Chat window in n8n (bottom right).
3. Type something messy: "I loved the beer tour but the museum was closed and I'm very disappointed because I drove 2 hours to get here!"
4. Watch n8n reply: It will instantly say something like: "Sentiment: Negative. Action Item: Offer a 20% refund or a free ticket for the museum next time."

Exercise 1 - Q&A

5 min

Keyword of this approach

- **Efficiency**: How much time would this save a Marketing team handling 50 surveys?
- **Consistency**: Why is it better to use an n8n workflow instead of 10 different managers writing 10 different prompts?
- **Scalability**: What happens if we change input like a file excel? Hint: Start the workflow and still works again!

Why Orchestrate, Not Just Prompt?

From one-shot questions to repeatable data intelligence

Simple Prompting (claude, chatgpt, etc..)

- One question, one answer — then lost
- No memory between sessions (only compressed one that lose informations)
- Copy-paste data every time
- Output format changes randomly (one single word can change output)
- Hard to reproduce or share
- Works for 1 file, breaks at 10

Orchestration (n8n)

- Reusable workflow — run it 100 times
- Data flows automatically from file to AI
- Consistent, structured output every run
- Swap data source, keep the logic
- Shareable with your team instantly
- Scales from 1 file to 100 files

Prompt Engineering vs Prompt Orchestration

What's the Difference?

Prompt Engineering

- Writing carefully designed prompts to guide LLMs
- Optimizes wording, tone, role, and constraints

Works well for:

- Q&A
- Summaries
- Formatting (tables, bullet points, JSON)
- Limitation → Breaks in complex, multi-step reasoning

Prompt Orchestration

A "system-level" approach that chains prompts, tools, memory, and integrations combines (like APIs):

- Prompt Templates → Predefined task instructions
- Context Injection → Feed external data (retrieval, DB, APIs)
- Agents/Workflows → Decision-making across multiple steps
- Evaluation/Verification → Ensure correctness & reliability

Works well for:

- Multi-document summarization
- Conversational agents with memory
- Complex pipelines (search → reasoning → reporting)

Orchestration

Symply, governance and control

Workflow Components

NODE

Processes data (Code, Set, HTTP Request)

Code
Run custom JavaScript or Python code

Data table
Permanently save data across workflow executions in a table

HTTP Request
Makes an HTTP request and returns the response data

Webhook
Starts the workflow when a webhook is called

Combine items

Aggregate
Combine a field from many items into a list in a single item

Merge
Merges data of multiple streams once data from both is available

Summarize
Sum, count, max, etc. across items

Add or remove items

Filter
Keep only items matching a condition

Limit
Restrict the number of items

Remove Duplicates
Delete items with matching field values

Split Out
Turn a list inside item(s) into separate items

Convert data

Compression
Compress and decompress files

Convert to File
Convert JSON data to binary data

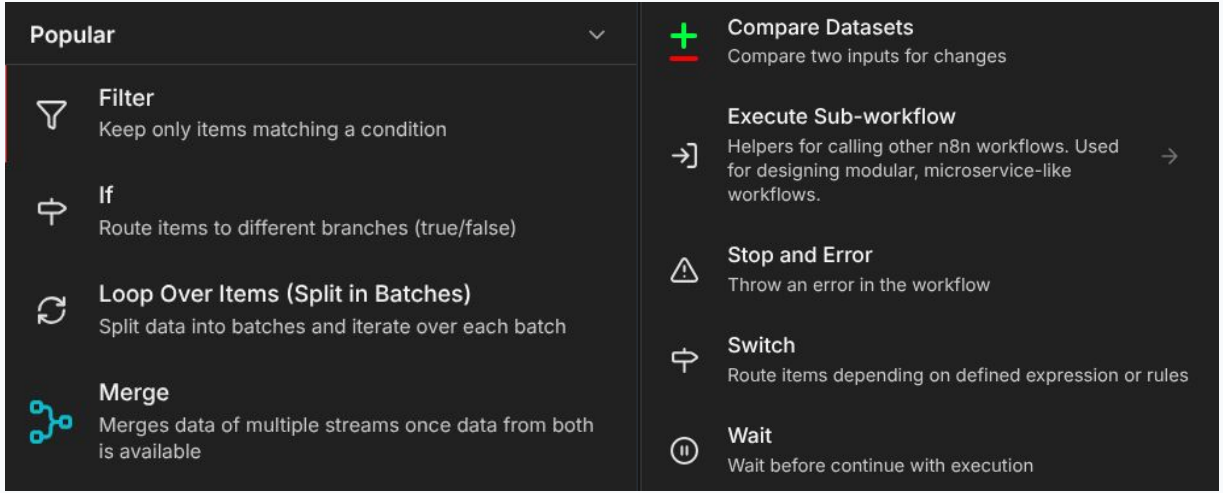
Crypto
Provide cryptographic utilities

Edit Image
Edits an image like blur, resize or adding border and text

Workflow Components

NODE

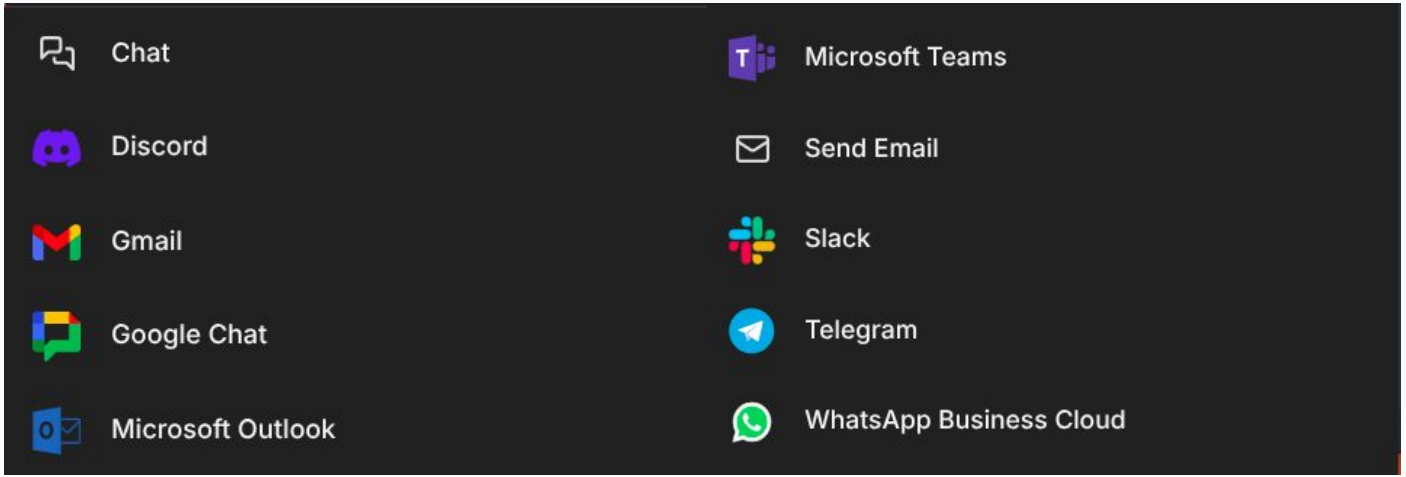
Conditional logic (IF, Switch, Merge)



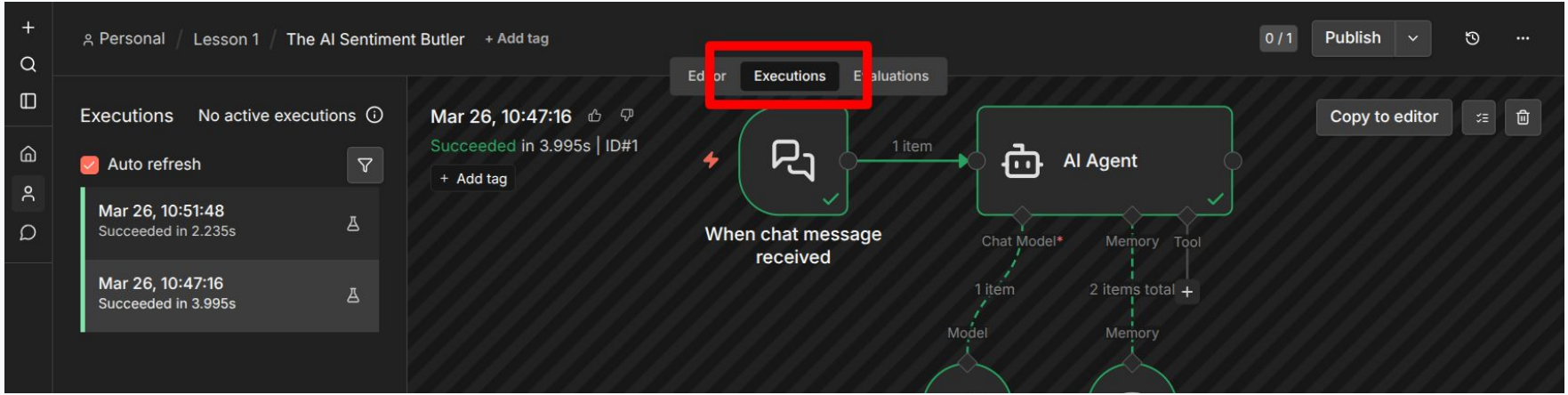
Workflow Components

OUTPUT

Final result
(File, Email,
Webhook, Slack)



Debug Workflow



Exercise 2
“The Market Opportunity Scout”

30-45 min

- Create a new Workflow named “The Market Opportunity Scout”
- **Objective**: Automate the transition from Raw Regional Data to a Strategic Investment Recommendation for the UK market.
- **The Scenario**: You are the International Expansion Manager for a winery. You have a budget to invest in the UK market, but you don't know where to put the money. Should you invest in "Digital Marketing," "Tasting Events," or "Sustainability Certifications"?

  Instead of reading the survey manually, you will build an n8n workflow that reads the UK data and gives you a $100,000 Investment Plan

Exercise 2
“The Market Opportunity Scout”

The Data Input (The "Eyes")

- Trigger: Add a Manual Trigger.
- Read File: Download with Google Drive: Regno_Unito_Generale.xlsx
- Transform: Add Extract from File (Excel). (Ensure "Header Row" is checked).
- Concept: You are teaching n8n to "look" at the UK market survey.

The Strategic Filter (The "Expert")

- Add AI Agent: Connect it to the Excel node.
- System Message: This is the most important part. Give the AI a "Management Persona":
    - "You are a Senior Investment Consultant. Analyze the UK survey data. Your goal is to identify the 3 highest-potential growth areas. Follow Pezzano's Step 3 (Largest Gaps) to see which behaviors are
- most unique to the UK audience compared to general trends."
- The Specific Prompt:
    - "Based on the attached data, create a $100,000 Investment Plan. Allocate the budget across 3 specific initiatives and justify each choice using a statistic from the table."

Comments

- Add "Stop and Error" (Optional but good): Explain to students that if the file is missing, the manager should get a clear alert.
- Add "Chat Publicly Available” in order to have a specific ss

Exercise 2 - Q&A

5 min

Some tricks:

- **Streaming view**
  enable on chat view and ai agent view seems to be more ChatGPT feel

- **Reduce number of attachment rows**
  This will redude input token and also AI costs and output timing

- **Permit uploaded files**
  In this case we can change file in every chat prompt

- **Chat Memory**
  This trick enable you to improve the chat session with multiple documents and data

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End of Lesson 1

Exercise 3
“Market Comparator”

Concept

A 5-node workflow where the student reads a specific country survey data (es. Dati_Birra_IT_20260217.xlsx), sends it to an AI agent that produces a structured "Market Snapshot Card" — a fixed-format mini-report with exactly 3 sections. Focus to control AI output format through prompt design.

Key point

- Simple: only 5 nodes, linear flow (no branching)
- New skill: focuses on output formatting via system prompt (the existing exercises focus on analysis methodology but don't teach structured output)
- Uses real data from "Dati Grezzi" (the Italian beer file, which none of the existing exercises use directly)
- Different from existing exercises: the Prompt Orchestrator is open-ended; this one forces a rigid card format — a key real-world skill

Workflow structure

- Create Manual Trigger
- Create Download file (or extract file passed via chat)
- Extract from File survey raw data
- Create AI Agent
- Create Chat Response

Specific output

You must write a system prompt that forces the AI to output exactly this format:

📊 MARKET SNAPSHOT: [Country]

🔍 TOP FINDING: [one sentence — the single biggest insight]
📈 STRONGEST SEGMENT: [which group shows highest values and in what]
⚠️ WATCH OUT: [one surprising or counter-intuitive pattern]